[1]:
```python
import subprocess

# Launch R script in background, detached
proc = subprocess.Popen(
    ["conda", "run", "-n", "RtorchGIS", "Rscript", "shiny-server.R"]
)
```

[9]:
```python
import os
prefix = os.environ.get("JUPYTERHUB_SERVICE_PREFIX", "/")

from IPython.display import IFrame
IFrame(f"{prefix}proxy/8050/", width="100%", height="1250")
```

[9]:

NN vs PINN — Sparse Seasonal Signal (R/torch)

Signal & data

Seasonal period (days)

60 180 365

60 95 130 165 200 235 270 305 340 365

Damping α (1/day)

0.05 0.2 0.6

0.05 0.11 0.17 0.23 0.29 0.35 0.41 0.47 0.53 0.59 0.6

Forcing β

0.2 1 2

0.2 0.4 0.6 0.8 1 1.2 1.4 1.6 1.8 2

Number of observations

8 20 60

8 14 20 26 32 38 44 50 56 60

Obs noise SD

0 2.2 3.6

0 0.36 0.72 1.08 1.44 1.8 2.16 2.52 2.88 3.24 3.6

Seed

1

Training

Epochs

100 600 2,000

100 300 500 700 900 1,100 1,300 1,500 1,700 1,900 2,000

Learning rate

0.001

Harmonic order H (features)

1 2 4

1 2 3 4

PINN physics weight λ

0 1 10

0 1 2 3 4 5 6 7 8 9 10

Generate data & train

Tip: increase λ to enforce physics more strongly, or increase epochs.

Fit | About

Plain NN (blue) vs PINN (red) on sparse seasonal data